Ellipse

Mind map

Recognition

- $\Delta \neq 0,\ h^2 < ab$
- $PS_1 + PS_2 = 2a$

Standard ellipse

- $b^2 = a^2(1 - e^2)$
- $S(\pm ae, 0),\ x = \pm\frac{a}{e}$
- LR $\frac{2b^2}{a}$

Focal distance

- $a \mp ex_1$
- Sum $2a$

Auxiliary circle

- $x^2 + y^2 = a^2$
- $(a\cos\theta, b\sin\theta)$

Shifted / other forms

- Complete squares
- $PS = e\,PM$
- $\frac{p_2^2}{a^2} + \frac{p_1^2}{b^2} = 1$

Position, chord

- $S_1 \lessgtr 0$
- Focal chord $\frac{e-1}{e+1}$

Tangency

- $c^2 = a^2m^2 + b^2$
- Contact $\left(-\frac{a^2m}{c}, \frac{b^2}{c}\right)$

Director circle

- $x^2 + y^2 = a^2 + b^2$
- Use $m_1 + m_2,\ m_1m_2$

Tangent

- $T = 0$
- $\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1$
- Least area ab

Normal

- $ax\sec\theta - by\operatorname{cosec}\theta = a^2 - b^2$
- Up to 4 normals

Minimum distance

- Parallel tangent
- Along the normal

Chords

- $T = S_1$
- $T = 0,\ SS_1 = T^2$

Pole, diameter

- $\left(-\frac{a^2l}{n}, -\frac{b^2m}{n}\right)$
- $m_1m_2 = -\frac{b^2}{a^2}$

Highlights

- $p_1p_2 = b^2$
- Feet on aux. circle
- Reflection via foci